

Pregame talks

Math AA HL

Sprint 2 · Sections 2.11 and 2.12, Polynomials

A root is a factor in disguise

Time limit: 20 minutes, strict. Start a timer. Stop when it rings, finished or not. Answers are on page 2: use them if you are stuck, and say so in the email. The point is to try.

What to send, by Thursday 8 October, 20:00: one email to Gauti.Eggertsson@gmail.com, subject “Sprint 2”, three lines:

Done at [time]. Got to problem [n]. Looked at answers for [which, or none].

On the weekend you redo these with Gauti over Zoom, without the answers.

1. $f(x) = x^3 + x^2 - x + 2$. Find the remainder when $f(x)$ is divided by $(x - 1)$, by $(x + 1)$, and by $(x + 2)$. Which one is a factor?

2. $f(x) = 2x^3 - 12x^2 + 22x - 12$. (a) Show that $x = 1$ is a zero. (b) Divide by $(x - 1)$ to get a quadratic. (c) Find all three zeros and write $f(x) = a(x - p)(x - q)(x - r)$.

3. $f(x) = x^3 + 2x^2 - 5x + k$. Find k if $(x - 2)$ is a factor of $f(x)$.

4. Vieta. (a) For $f(x) = 2x^3 - 12x^2 + 22x - 12$ (roots 1, 2, 3) check the sum and product of the roots against the coefficients. (b) The cubic $ax^3 + 6x^2 - 8x + 9$ has three real roots whose sum is 2. Find a and the product of the roots.

Look only when stuck, and say which ones in the email. On the weekend these are redone without this page.

1. Remainder theorem: $f(1) = 3$, $f(-1) = -1 + 1 + 1 + 2 = 3$, $f(-2) = -8 + 4 + 2 + 2 = 0$. So $(x + 2)$ is a factor. (Dividing by $x + 2$ means evaluating at -2 .)
2. (a) $f(1) = 2 - 12 + 22 - 12 = 0$. (b) $f(x) = (x - 1)(2x^2 - 10x + 12)$. (c) $2x^2 - 10x + 12 = 2(x - 2)(x - 3)$, so the zeros are 1, 2, 3 and $f(x) = 2(x - 1)(x - 2)(x - 3)$. Do not lose the leading 2.
3. Factor theorem: $f(2) = 0$. $8 + 8 - 10 + k = 6 + k = 0$, so $k = -6$.
4. (a) Sum = $6 = -\frac{-12}{2}$; product = $6 = (-1)^3 \cdot \frac{-12}{2}$. (b) Sum = $-\frac{6}{a} = 2$, so $a = -3$. Product = $(-1)^3 \cdot \frac{9}{a} = -\frac{9}{-3} = 3$.